F78PA Probability & Statistics A

Tutorial 4 – Solutions

1. Let X denote the number of sixes obtained in two rolls of a fair die.

   Then $X \sim \text{Bin}(2, \frac{1}{6})$.

   The net profit W is given by the winnings minus the £1 paid to play the game, so the probability mass function of W is

$$\begin{aligned} p_W(9) = \mathbb{P}(W=9) &= \mathbb{P}(X=2) = \binom{2}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^0 = \frac{1}{36} \\ p_W(1) = \mathbb{P}(W=1) &= \mathbb{P}(X=1) = \binom{2}{1}\left(\frac{1}{6}\right)^1\left(\frac{5}{6}\right)^1 = \frac{10}{36} \\ p_W(-1) = \mathbb{P}(W=-1) &= \mathbb{P}(X=0) = \binom{2}{0}\left(\frac{1}{6}\right)^0\left(\frac{5}{6}\right)^2 = \frac{25}{36} \end{aligned}$$

   with $p_W(w) = 0$ for $w \notin \{9, 1, -1\}$.

   Expected net winnings

$$\mathbb{E}(W) = 9 \times \frac{1}{36} + 1 \times \frac{10}{36} - 1 \times \frac{25}{36} = -\frac{6}{36} = -\frac{1}{6} \approx -0.1667$$

   If the player pays £c to play the game, then the probability mass function of W becomes

$$p_W(w) = \begin{cases} \frac{1}{36} & w = 10 - c \\ \frac{10}{36} & w = 2 - c \\ \frac{25}{36} & w = -c \end{cases}$$

   with expected net profit

$$\mathbb{E}(W) = (10-c) \times \frac{1}{36} + (2-c) \times \frac{10}{36} - c \times \frac{25}{36} = \frac{5}{6} - c$$

   So the player's expected net profit $\mathbb{E}(W)$ is zero if $c = \frac{5}{6}$.

2. Using formula for expectation of a function of a random variable,

$$\begin{aligned} \mathbb{E}(S) = \mathbb{E}(\ln(R)) &= \sum_r \ln(r) p_R(r) \\ &= 0.1 \times \ln(1) + 0.1 \times \ln(2) + 0.3 \times \ln(3) + 0.3 \times \ln(4) + 0.2 \times \ln(5) \\ &\approx 1.1367 \end{aligned}$$

   To find variance, first

$$\begin{aligned} \mathbb{E}(S^2) = \mathbb{E}((\ln(R))^2) &= \sum_r (\ln(r))^2 p_R(r) \\ &= 0.1(\ln(1))^2 + 0.1(\ln(2))^2 + 0.3(\ln(3))^2 + 0.3(\ln(4))^2 + 0.2(\ln(5))^2 \\ &\approx 1.5047 \end{aligned}$$

   and then

$$\mathbb{V}\text{ar}(S) = \mathbb{E}(S^2) - (\mathbb{E}(S))^2 \approx 0.2127$$

3. For $X \sim \text{Bin}(n,p)$,

$$\begin{aligned}
\mathbb{E}(X) &= \sum_{k=0}^{n} k\binom{n}{k}p^k(1-p)^{n-k} \\
&= \sum_{k=0}^{n} k\,\frac{n!}{k!\,(n-k)!}p^k(1-p)^{n-k} \\
&= \sum_{k=1}^{n} k\,\frac{n!}{k!\,(n-k)!}p^k(1-p)^{n-k} \\
&= \sum_{k=1}^{n} \frac{n!}{(k-1)!\,(n-k)!}p^k(1-p)^{n-k} \\
&= np\sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!\,(n-k)!}p^{k-1}(1-p)^{n-k} \\
&= np\sum_{j=0}^{n-1} \binom{n-1}{j}p^j(1-p)^{(n-1)-j} \text{ where } j=k-1 \\
&= np\sum_{j=0}^{n-1} \mathbb{P}(Y=j) \text{ where } Y \sim \text{Bin}(n-1,p)
\end{aligned}$$

Now the probability mass function of $Y \sim \text{Bin}(n-1,p)$ sums to 1, which is to say that $\sum_{j=0}^{n-1}\mathbb{P}(Y=j)=1$, and so we have shown that $\mathbb{E}(X)=np$ as required.

To find the variance, first

$$\begin{aligned}
\mathbb{E}(X(X-1)) &= \sum_{k=0}^{n} k(k-1)\binom{n}{k}p^k(1-p)^{n-k} \\
&= \sum_{k=0}^{n} k(k-1)\,\frac{n!}{k!\,(n-k)!}p^k(1-p)^{n-k} \\
&= \sum_{k=2}^{n} k(k-1)\,\frac{n!}{k!\,(n-k)!}p^k(1-p)^{n-k} \\
&= \sum_{k=2}^{n} \frac{n!}{(k-2)!\,(n-k)!}p^k(1-p)^{n-k} \\
&= n(n-1)p^2\sum_{k=2}^{n} \frac{(n-2)!}{(k-2)!\,(n-k)!}p^{k-2}(1-p)^{n-k} \\
&= n(n-1)p^2\sum_{i=0}^{n-2} \binom{n-2}{i}p^i(1-p)^{(n-2)-i} \text{ where } i=k-2 \\
&= n(n-1)p^2\sum_{i=0}^{n-2} \mathbb{P}(V=i) \text{ where } V \sim \text{Bin}(n-2,p) \\
&= n(n-1)p^2
\end{aligned}$$

using the fact that the probability mass function of $V \sim \text{Bin}(n-2,p)$ sums to 1. Notice that the above argument is only valid provided $n \geq 2$, but for the case $n=1$ it is certain that either $X=0$ or $X=1$, so we have $E(X(X-1))=0$, and the formula $\mathbb{E}(X(X-1))=n(n-1)p^2$ remains true.

Now

$$\begin{aligned}\mathbb{V}\mathrm{ar}(X) &= E(X^2) - (E(X))^2 \\ &= E(X(X-1)+X) - (E(X))^2 \\ &= E(X(X-1)) + E(X) - (E(X))^2 \\ &= n(n-1)p^2 + np - (np)^2 \\ &= np(1-p)\end{aligned}$$

as required.

4. X is a random variable with $\mathbb{E}[X] = \frac{3}{5}$ and with density

$$f(x) = \begin{cases} a+bx & 0 < x < 1, \\ 0 & \text{otherwise,} \end{cases}$$

(a) To determine the values of the two unknowns a and b, we need two equations that they satisfy.
Since $f(x)$ is a density, we have

$$1 = \int_{x=-\infty}^{\infty} f(x)\,dx = \int_{x=0}^{1} (a+bx)\,dx = \left[ax + \frac{bx^2}{2}\right]_{x=0}^{1} = a + \frac{b}{2}$$

Since $\mathbb{E}[X] = \frac{3}{5}$, we have

$$\frac{3}{5} = \int_{x=-\infty}^{\infty} xf(x)\,dx = \int_{x=0}^{1} x(a+bx)\,dx = \left[\frac{ax^2}{2} + \frac{bx^3}{3}\right]_{x=0}^{1} = \frac{a}{2} + \frac{b}{3}$$

We now solve the system of equations

$$\begin{aligned} a + \frac{b}{2} &= 1 \\ \frac{a}{2} + \frac{b}{3} &= \frac{3}{5} \end{aligned}$$

and find that $a = \frac{2}{5}$, $b = \frac{6}{5}$.

(b)

$$\begin{aligned}\mathbb{P}(1/4 \le X \le 1/2) = \int_{x=1/4}^{1/2} f(x)\,dx &= \int_{x=1/4}^{1/2} \frac{2}{5} + \frac{6x}{5}\,dx \\ &= \left[\frac{2x}{5} + \frac{3x^2}{5}\right]_{x=1/4}^{1/2} = \frac{17}{80} = 0.2125\end{aligned}$$

(c) To find $\mathbb{V}\mathrm{ar}[X]$, first

$$\begin{aligned}E[X^2] = \int_{x=-\infty}^{\infty} x^2 f(x)\,dx &= \int_{x=0}^{1} x^2\left(\frac{2}{5} + \frac{6x}{5}\right)dx \\ &= \left[\frac{2x^3}{15} + \frac{3x^4}{10}\right]_{x=0}^{1} = \frac{13}{30}\end{aligned}$$

and now $\mathbb{V}\mathrm{ar}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{13}{30} - (\frac{3}{5})^2 = \frac{11}{150} \approx 0.0733$.

(d) With $Y = X^2$,

$$\begin{aligned}\mathbb{P}(1/4 \le Y \le 1/2) = \mathbb{P}(1/4 \le X^2 \le 1/2) &= \mathbb{P}(1/2 \le X \le 1/\sqrt{2}) \\ &= \int_{x=1/2}^{1/\sqrt{2}} \frac{2}{5} + \frac{6x}{5}\,dx \\ &= \left[\frac{2x}{5} + \frac{3x^2}{5}\right]_{x=1/2}^{1/\sqrt{2}} \\ &= \frac{\sqrt{2}}{5} - \frac{1}{20} \approx 0.2328\end{aligned}$$

(e) To find $\mathbb{V}\text{ar}[Y]$, first notice that $\mathbb{E}[Y] = \mathbb{E}[X^2] = \frac{13}{30}$ from part (c) above. Next,

$$E[Y^2] = E[X^4] = \int_{x=-\infty}^{\infty} x^4 f(x)\,dx = \int_{x=0}^{1} x^4 \left(\frac{2}{5} + \frac{6x}{5}\right) dx$$
$$= \left[\frac{2x^5}{25} + \frac{x^6}{5}\right]_{x=0}^{1} = \frac{7}{25}$$

and now $\mathbb{V}\text{ar}[Y] = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = \frac{7}{25} - (\frac{13}{30})^2 = \frac{83}{900} \approx 0.0922$.

5. Weekly demand X has probability density function

$$f(x) = \begin{cases} 5(1-x)^4 & 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Denoting by t the size of the station's storage tank, then the probability that the station runs out of petrol during the week is $\mathbb{P}(X > t)$.

Now for $0 \le t \le 1$,

$$\mathbb{P}(X > t) = \int_{x=t}^{\infty} f(x)\,dx = \int_{x=t}^{1} 5(1-x)^4\,dx$$
$$= \left[-(1-x)^5\right]_{x=t}^{1} = (1-t)^5$$

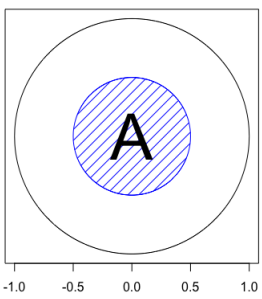
We are asked to find the value t such that $\mathbb{P}(X > t) = 0.01$; that is,

$$(1-t)^5 = 0.01 \quad \Rightarrow \quad 1-t = 0.01^{1/5} \quad \Rightarrow \quad t = 1 - 0.01^{1/5} \approx 0.6019 \text{ (thousand litres)}$$

In order that the probability that the station runs out of petrol during the week is 0.01, the size of the station's storage tank must be (approximately) 602 litres.

6. (a) Since the probability that the arrow hits a region A is proportional to the area of A, we have

$$\mathbb{P}(\text{arrow lands closer to the centre than the edge}) = \mathbb{P}(\text{arrow lands in shaded region})$$
$$= \frac{\text{Area of shaded region}}{\text{Area of target}}$$
$$= \frac{\pi \times (1/2)^2}{\pi \times 1^2} = \frac{1}{4}$$



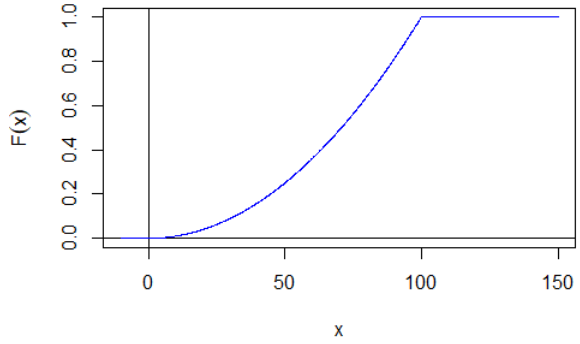
(b) X = distance (cm) between where the arrow hits and the centre of the target.
So $\text{Im}(X) = [0, 100]$ (the possible values of X)
For $x < 0$, $F_X(x) = \mathbb{P}(X \le x) = 0$.
For $x \ge 100$, $F_X(x) = \mathbb{P}(X \le x) = 1$.

For $0 \leq x \leq 100$,

$$\begin{aligned} F_X(x) = \mathbb{P}(X \leq x) &= \frac{\text{Area of disk with radius } x}{\text{Area of target}} \\ &= \frac{\pi x^2}{\pi \times (100)^2} \text{ (since measurements in cm)} \\ &= \frac{x^2}{10000} \end{aligned}$$

That is,

$$F_X(x) = \begin{cases} 0 & x < 0, \\ \frac{x^2}{10000} & 0 \leq x \leq 100, \\ 1 & 100 < x. \end{cases}$$



Notice that F_X is a continuous function.
$\mathbb{P}(30 < X < 90) = \mathbb{P}(30 < X \leq 90) = F_X(90) - F_X(30) = 18/25$ (where the first equality makes use of the fact that $\mathbb{P}(X = 90) = 0$ because F_X is continuous)
To find $\mathbb{E}(X)$, we first calculate the probability density function $f_X(x)$, obtained by differentiating $F_X(x)$, so

$$f_X(x) = \begin{cases} \frac{x}{5000} & 0 \leq x \leq 100, \\ 0 & \text{otherwise.} \end{cases}$$

Now

$$\mathbb{E}(X) = \int_{x=0}^{100} x \left(\frac{x}{5000}\right) dx = \left[\frac{x^3}{15000}\right]_{x=0}^{100} = \frac{100^3}{15000} - 0 = \frac{200}{3} \approx 66.67$$

This seems reasonable, as we can see from the graph of $F_X(x)$ that X is more likely to be greater than 50 than less than 50.

7. Suppose X is a random variable with density

$$f_X(x) = \begin{cases} ax^2 & 0 \leq x \leq 4, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Since $f(x)$ is a density, we require that $\int_{x=-\infty}^{\infty} f(x)\,dx = 1$. Now $\int_{x=-\infty}^{\infty} f(x)\,dx = \int_{x=0}^{4} ax^2\,dx = [\frac{a}{3}x^3]_{x=0}^{4} = \frac{64a}{3}$, and so $a = 3/64$.

(b) Let $Y = 1/\sqrt{X}$.

To determine $\text{Im}(Y)$, first note that $\text{Im}(X) = \{x : f(x) > 0\} = (0, 4]$, so that $0 < X \leq 4$, so $0 < \sqrt{X} \leq 2$, and hence $\frac{1}{2} \leq 1/\sqrt{X} < \infty$. That is, $\text{Im}(Y) = [\frac{1}{2}, \infty)$.

To find the density function of Y, first note that since $\text{Im}(Y) = [\frac{1}{2}, \infty)$, then $f_Y(y) = 0$ for $y < \frac{1}{2}$. For $y \geq \frac{1}{2}$,

$$\begin{aligned} F_Y(y) = \mathbb{P}(Y \leq y) &= \mathbb{P}\left(\frac{1}{\sqrt{X}} \leq y\right) \\ &= \mathbb{P}\left(\sqrt{X} \geq \frac{1}{y}\right) \\ &= \mathbb{P}\left(X \geq \frac{1}{y^2}\right) \\ &= 1 - \mathbb{P}\left(X < \frac{1}{y^2}\right) \\ &= 1 - F_X\left(\frac{1}{y^2}\right) \end{aligned}$$

So now for $y \geq 1/2$,

$$\begin{aligned} f_Y(y) = F'_Y(y) &= \frac{d}{dy}\left(1 - F_X\left(\frac{1}{y^2}\right)\right) \\ &= \frac{2}{y^3} F'_X\left(\frac{1}{y^2}\right) \quad \text{(chain rule)} \\ &= \frac{2}{y^3} f_X\left(\frac{1}{y^2}\right) \\ &= \frac{2}{y^3}\left(\frac{3}{64}\frac{1}{y^4}\right) = \frac{3}{32y^7} \end{aligned}$$

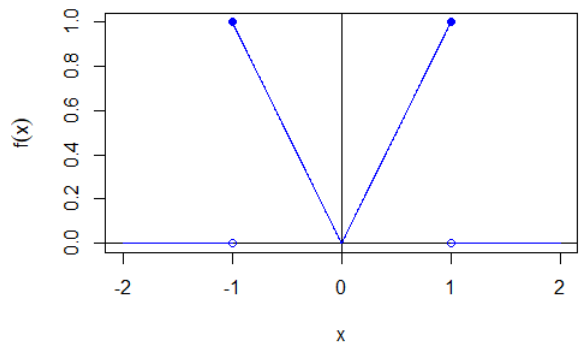
That is,

$$f_Y(x) = \begin{cases} \frac{3}{32y^7} & y \geq 1/2, \\ 0 & \text{otherwise.} \end{cases}$$

8. Random variable X has density

$$f_X(x) = \begin{cases} |x| & -1 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Sketch:



$$\begin{aligned}\mathbb{E}(X) = \int_{\mathbb{R}} x f_X(x)\,dx &= \int_{x=-1}^{1} x\,|x|\,dx \\ &= \int_{x=-1}^{0} -x^2\,dx + \int_{x=0}^{1} x^2\,dx \\ &= \left[-\frac{x^3}{3}\right]_{x=-1}^{0} + \left[\frac{x^3}{3}\right]_{x=0}^{1} \\ &= -\frac{1}{3} + \frac{1}{3} = 0\end{aligned}$$

This is unsurprising, since the density $f_X(x)$ is symmetric about $x = 0$, so $\mathbb{E}(X) = 0$.

To find $\mathbb{V}\mathrm{ar}[X]$,

$$\begin{aligned}\mathbb{E}(X^2) = \int_{\mathbb{R}} x^2 f_X(x)\,dx &= \int_{x=-1}^{1} x^2\,|x|\,dx \\ &= \int_{x=1}^{0} -x^3\,dx + \int_{x=0}^{1} x^3\,dx \\ &= \left[-\frac{x^4}{4}\right]_{x=-1}^{0} + \left[\frac{x^4}{4}\right]_{x=0}^{1} \\ &= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}\end{aligned}$$

so $\mathbb{V}\mathrm{ar}(X) = \frac{1}{2} - 0^2 = \frac{1}{2}$

(b) For $Y = |X|$,

$$\begin{aligned}\mathbb{E}[Y] &= \mathbb{E}[|X|] \\ &= \int_{\mathbb{R}} |x| f_X(x)\,\mathrm{d}x \\ &= \int_{-1}^{1} |x||x|\,\mathrm{d}x \\ &= \int_{-1}^{1} x^2\,\mathrm{d}x \\ &= \left[\frac{1}{3}x^3\right]_{x=-1}^{1} \\ &= \frac{1}{3}\left(1^3 - (-1)^3\right) = \frac{2}{3}.\end{aligned}$$

Note that $Y^2 = |X|^2 = X^2$ so $\mathbb{E}[Y^2] = \mathbb{E}[X^2] = \frac{1}{2}$ from part (a), therefore

$$\mathbb{V}\mathrm{ar}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}.$$

9. (a) $\mathbb{E}[X] = \int_{x=0}^{3} x\left(x^2/9\right)dx = \left[x^4/36\right]_{x=0}^{3} = (3^4/36) - 0 = 9/4$

   (b) Since $Y = 1/X$, to compute $\mathbb{E}[Y]$ use the fact that $\mathbb{E}(g(X)) = \int_{\mathbb{R}} g(x) f_X(x)\,dx$. $\mathbb{E}[Y] = \mathbb{E}[1/X] = \int_{x=0}^{3}(1/x)\left(x^2/9\right)dx = \int_{x=0}^{3}(x/9)\,dx = \left[x^2/18\right]_{x=0}^{3} = (3^2/18) - 0 = 1/2$

10. Suppose that $X \sim \mathrm{Exp}(\lambda)$, with $\lambda > 0$. Then X has density function

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

(a) $\mathbb{V}\text{ar}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

Now

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f_X(x)\, dx \quad = \quad \int_{x=0}^{\infty} x(\lambda e^{-\lambda x})\, dx$$

Integrate by parts, with

$$\begin{aligned} & u = x, \qquad \frac{dv}{dx} = \lambda e^{-\lambda x}, \\ \Rightarrow \qquad & \frac{du}{dx} = 1, \qquad v = -e^{-\lambda x} \end{aligned}$$

so

$$\begin{aligned} \mathbb{E}[X] &= \int_{x=0}^{\infty} x(\lambda e^{-\lambda x})\, dx \\ &= \left[-xe^{-\lambda x}\right]_{x=0}^{\infty} + \int_{x=0}^{\infty} e^{-\lambda x}\, dx \\ &= 0 + \left[\frac{e^{-\lambda x}}{-\lambda}\right]_{x=0}^{\infty} \\ &= \frac{1}{\lambda} \end{aligned}$$

where, in evaluating the upper end points of the integrals, we have made use of the fact that $\lim_{x\to\infty} e^{-\lambda x} = \lim_{x\to\infty} xe^{-\lambda x} = 0$, for any $\lambda > 0$.

Next,

$$\mathbb{E}[X^2] = \int_{\mathbb{R}} x^2 f_X(x)\, dx \quad = \quad \int_{x=0}^{\infty} x^2(\lambda e^{-\lambda x})\, dx$$

Integrate by parts, with

$$\begin{aligned} & u = x^2, \qquad \frac{dv}{dx} = \lambda e^{-\lambda x}, \\ \Rightarrow \qquad & \frac{du}{dx} = 2x, \qquad v = -e^{-\lambda x} \end{aligned}$$

so

$$\begin{aligned} \mathbb{E}[X^2] &= \int_{x=0}^{\infty} x^2(\lambda e^{-\lambda x})\, dx \\ &= \left[-x^2 e^{-\lambda x}\right]_{x=0}^{\infty} + \int_{x=0}^{\infty} 2xe^{-\lambda x}\, dx \\ &= 0 + 2\int_{x=0}^{\infty} xe^{-\lambda x}\, dx \end{aligned}$$

where we have made use of the fact that $\lim_{x\to\infty} x^2 e^{-\lambda x} = 0$, for any $\lambda > 0$. The remaining integral is one that we've already evaluated when working out $\mathbb{E}[X]$, when we found $\mathbb{E}[X] = \int_{x=0}^{\infty} \lambda xe^{-\lambda x}\, dx = 1/\lambda$, so that $\int_{x=0}^{\infty} xe^{-\lambda x}\, dx = 1/\lambda^2$, and hence

$$\mathbb{E}[X^2] \quad = \quad \frac{2}{\lambda^2}$$

Finally,

$$\mathbb{V}\text{ar}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2},$$

as required.

(b) Let $Y = cX$, where $X \sim \text{Exp}(\lambda)$ and $c > 0$.
Since $\text{Im}\, X = [0, \infty)$, then $\text{Im}\, Y = [0, \infty)$.
The density $f_Y(y)$ of Y therefore satisfies $f_Y(y) = 0$ for $y < 0$.
For $y \geq 0$, first consider the distribution function $F_Y(y)$.

$$\begin{aligned} F_Y(y) = \mathbb{P}(Y \leq y) &= \mathbb{P}(cX \leq y) \\ &= \mathbb{P}\left(X \leq \frac{y}{c}\right) \\ &= F_X\left(\frac{y}{c}\right) \end{aligned}$$

Differentiate, using the chain rule.

$$\begin{aligned} f_Y(y) = F_Y'(y) &= \frac{d}{dy}\left(F_X\left(\frac{y}{c}\right)\right) \\ &= \frac{1}{c} F_X'\left(\frac{y}{c}\right) \\ &= \frac{1}{c} f_X\left(\frac{y}{c}\right) \\ &= \frac{1}{c}\left(\lambda e^{-\lambda y/c}\right) \qquad \text{(for } y \geq 0\text{)} \\ &= (\lambda/c) e^{-(\lambda/c)y} \end{aligned}$$

So the density function of Y is

$$f_Y(y) = \begin{cases} (\lambda/c) e^{-(\lambda/c)y} & y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

This is the density function of the $\text{Exp}(\lambda/c)$ distribution, so $Y = cX$ is exponentially distributed with parameter λ/c.

Exam Style Question

(a) i) A sketch of f_X is given in Figure 1. [2 marks]

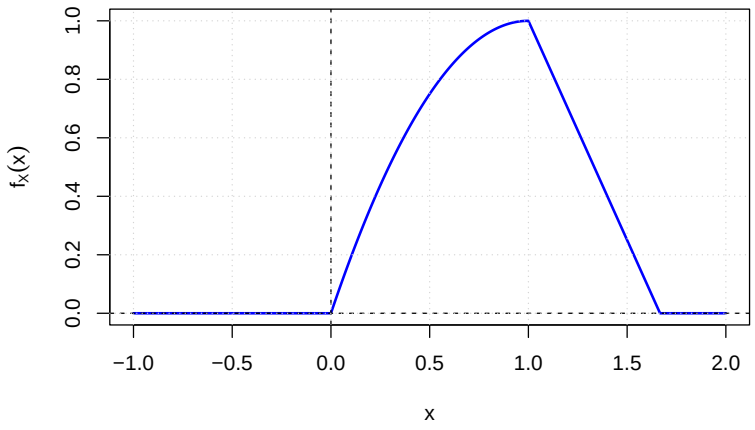


Figure 1: A plot of the function f_X.

We observe from the graph that $f_X(x) \geq 0$ for all $x \in \mathbb{R}$. We have

$$\begin{aligned}\int_{\mathbb{R}} f_X(x)\,\mathrm{d}x &= \int_0^1 (2x - x^2)\,\mathrm{d}x + \int_1^{\frac{5}{3}} \frac{5-3x}{2}\,\mathrm{d}x \\ &= \left[x^2 - \frac{x^3}{3}\right]_{x=0}^{1} + \left[\frac{5x}{2} - \frac{3x^2}{4}\right]_{x=1}^{\frac{5}{3}} \\ &= \left(1 - \frac{1}{3}\right) + \left(\frac{25}{6} - \frac{75}{36}\right) - \left(\frac{5}{2} - \frac{3}{4}\right) \\ &= \frac{24}{36} + \frac{75}{36} - \frac{63}{36} \\ &= 1.\end{aligned}$$

[1 mark]

Therefore, f_X is a valid probability density function because it is non-negative and the area under the graph is 1.

ii) The cumulative distribution function is $F_X(x) = 0$ for $x < 0$ and $F_X(x) = 1$ for $x > 5/3$. [1 mark]

For $0 \leq x \leq 1$,

$$\begin{aligned}F_X(x) &= \int_{-\infty}^{x} f(u)\,\mathrm{d}u \\ &= \int_0^1 (2u - u^2)\,\mathrm{d}u \\ &= \left[u^2 - \frac{u^3}{3}\right]_{u=0}^{x} \\ &= x^2 - \frac{x^3}{3}.\end{aligned}$$

[1 mark]

For $1 \leq x \leq 5/3$,

$$\begin{aligned}F_X(x) &= \int_{-\infty}^{x} f(u)\,\mathrm{d}u \\ &= (1)^2 - \frac{(1)^3}{3} + \int_1^x \frac{5-3u}{2}\,\mathrm{d}u \\ &= \frac{2}{3} + \left[\frac{5u}{2} - \frac{3u^2}{4}\right]_1^x \\ &= \frac{2}{3} + \left(\frac{5x}{2} - \frac{3x^2}{4}\right) - \frac{7}{4} \\ &= \frac{5x}{2} - \frac{3x^2}{4} - \frac{13}{12} = \frac{1}{12}\left(30x - 9x^2 - 13\right).\end{aligned}$$

[1 mark]

So the cumulative distribution function is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^2 - \frac{1}{3}x^3, & 0 \leq x < 1, \\ \frac{5}{2}x - \frac{3}{4}x^2 - \frac{13}{12}, & 1 \leq x < \frac{5}{3}, \\ 1 & x \geq \frac{5}{3}. \end{cases}$$

iii)

$$\begin{aligned}\mathbb{P}\left(X > \frac{3}{2} | X > 1\right) &= \frac{\mathbb{P}\left(\{X > \frac{3}{2}\} \cap \{X > 1\}\right)}{\mathbb{P}(X > 1)} \\ &= \frac{\mathbb{P}\left(X > \frac{3}{2}\right)}{\mathbb{P}(X > 1)} \\ &= \frac{1 - F_X(\frac{3}{2})}{1 - F_X(1)} \\ &= \frac{1 - \frac{47}{48}}{1 - \frac{2}{3}} \\ &= \frac{1}{16}.\end{aligned}$$

[2 marks]

iv)

$$\begin{aligned}\mathbb{E}[X] &= \int_{\mathbb{R}} x f_X(x)\,\mathrm{d}x \\ &= \int_0^1 x(2x - x^2)\,\mathrm{d}x + \int_1^{5/3} x\left(\frac{5 - 3x}{2}\right)\mathrm{d}x \\ &= \left[\frac{2}{3}x^3 - \frac{1}{4}x^4\right]_0^1 + \left[\frac{5}{4}x^2 - \frac{1}{2}x^3\right]_{x=1}^{5/3} \\ &= \frac{5}{12} + \left(\frac{125}{36} - \frac{125}{54}\right) - \frac{3}{4} \\ &= \frac{89}{108}.\end{aligned}$$

[2 marks]

(b) For $W \sim \mathrm{Exp}(2)$ we have $\mathbb{E}[W] = 1/2$ and $\mathbb{V}\mathrm{ar}(W) = 1/4$.

So

$$\mathbb{E}[Y] = \mathbb{E}[3W - 1] = 3\mathbb{E}[W] - 1 = \frac{3}{2} - 1 = \frac{1}{2}.$$

[1 mark]

and

$$\mathbb{V}\mathrm{ar}(Y) = \mathbb{V}\mathrm{ar}(3W - 1) = 3^2\mathbb{V}\mathrm{ar}(W) = \frac{9}{4}.$$

[1 mark]

[Total 12 marks]